18 Magnetic Fields

Magnets have fascinated people for centuries. Did you know that the magnetic compass has been used for over 2000 years? The earliest compasses were probably made from lodestone, which is a naturally occurring oxide of iron.



How does a compass show direction?

If you suspend a magnet horizontally from a piece of string, it will always rotate until it points in a north–south direction. The end of the magnet that points north is called the north-seeking pole or ***N-pole***. The other end is the south-seeking pole or ***S-pole***.

We have come a long way since the invention of the compass! Many everyday devices, from washing machines to computers and swipe-cards, depend on magnetism to work.

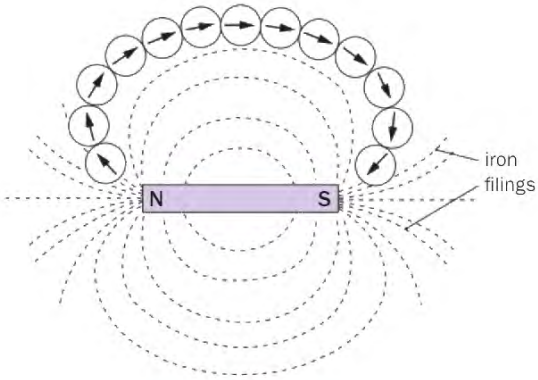
In this chapter you will learn:

- about magnetic field patterns and lines of flux,
- how to measure and calculate the strength of magnetic fields,
- how to calculate the force on a current in a magnetic field.

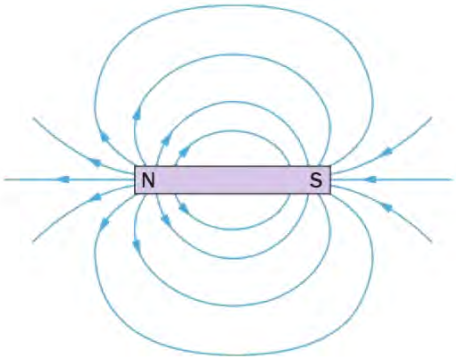
Magnetic fields

What happens if you bring the poles of two magnets together?

You will find that two N-poles repel, and two S-poles repel, but a N-pole and a S-pole attract. The effect of these forces can be felt even though the magnets are not in contact with each other. We say that a magnet has a **magnetic field** around it. The field is a region where the magnetic force can be felt.



How can we 'see the shape' of a magnetic field? If we place iron filings or plotting compasses in the field of the magnet, they will point along curved lines called **lines of flux**, as shown in the diagram:



The lines of flux show both the ***direction*** and the ***strength*** of the magnetic field. The direction of the line of flux at each point shows the direction of the force that a N-pole would feel if placed at that point in the field. This means that lines of flux always go from N-poles to S-poles.

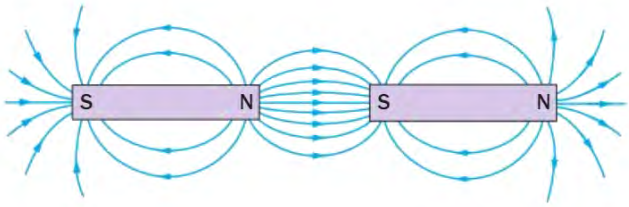
Where is the field strongest?

The more closely packed the lines of flux, the stronger the field. The field is strongest nearest the poles, as shown in the diagram:

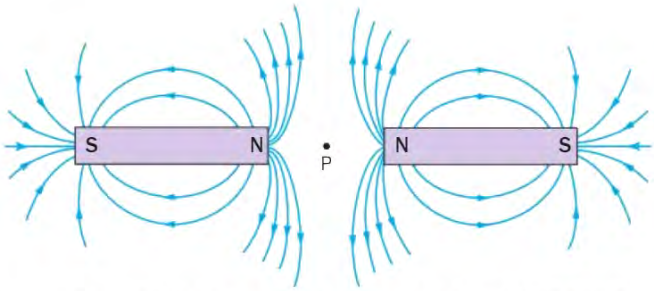
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More magnetic field patterns

Lines of flux can never cross: if they did it would mean that a compass would point in two directions! If two magnets are placed close to each other, the field produced is the result of the combined effect of both of the magnets:



There is an attraction between the N-pole and the S-pole.
Where is the field the strongest?



Can you see the repulsion between the two N-poles?
At point P the fields from the two magnets cancel out, and the resultant field strength is zero.
P is called a ***neutral point***.

Magnetic flux density

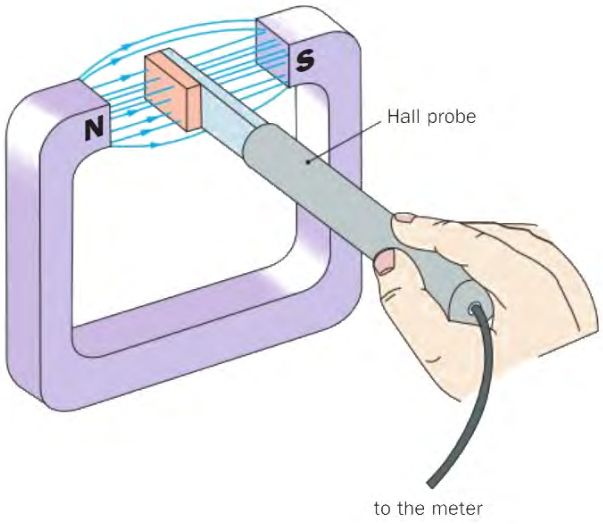
The stronger the magnetic field, the more densely packed the lines of flux. In fact, we describe the strength of the magnetic field by the **magnetic flux density, *B***.
This is the quantity of flux passing through unit area, at each point in the field. It is measured in ***tesla*** (T).

B is a vector quantity, because it has size and direction.
B is discussed in more detail on page 260.

How can we measure magnetic flux density?
A simple way is to use a **Hall probe** (see also page 267).
The diagram shows a Hall probe being used to measure the flux density between the poles of a horseshoe magnet:

The probe must be held in the field so that the lines of flux pass at right angles through its tip. The meter connected to the probe gives a direct reading of the flux density, in tesla.

The diagram shows how the flux passes through the tip of the Hall probe. Can you picture how the flux density will vary as the probe is moved to different positions in the magnetic field?

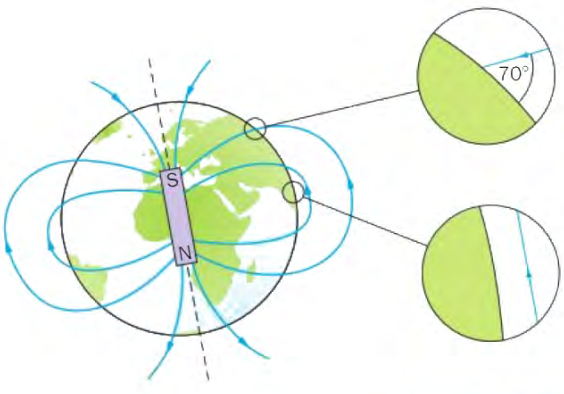


The Earth's magnetic field

Why does a compass line up in a north–south direction?
The Earth acts as if there is a huge bar magnet inside it.
In the diagram, notice that the S-pole of the imaginary magnet is in the northern hemisphere so as to attract the N-pole of a compass:

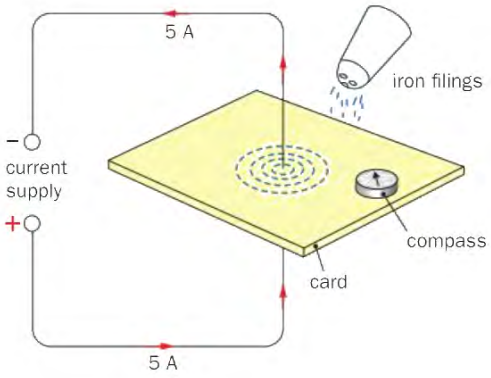
Can you also see from the diagram that the lines of flux act at an ***angle*** to the Earth's surface, except near the equator?
This angle is called the **angle of dip**. In the UK it is about 70°.

A compass needle held horizontally is only affected by the horizontal component of the magnetic flux. What do you think will happen to a compass needle at the magnetic north pole?



The magnetic effect of a current

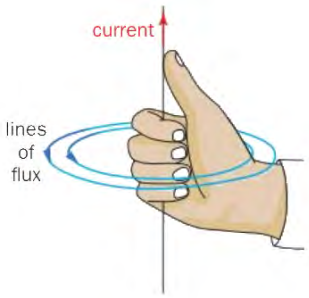
In 1819, Hans Christian Oersted noticed that a compass needle was deflected by an electric current in a nearby wire. In this way, he discovered the link between electricity and magnetism.
An electric current is always surrounded by a magnetic field.



The magnetic field around a long straight wire

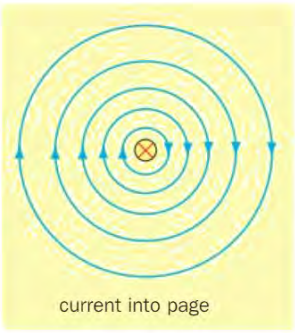
The diagram shows a wire carrying a current of about 5 amps.
If you sprinkle some iron filings on to the horizontal card and tap it gently, the iron filings will line up along the lines of flux as shown.
Can you see that the lines of flux are circles around the wire?

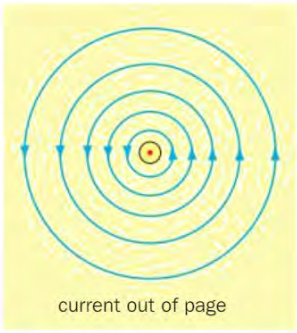
You can place a small compass on the card to find the direction of the magnetic field. With the current flowing up the wire, the compass will point anti-clockwise, as shown.
What will happen if you reverse the direction of the current?



The right-hand grip rule gives a simple way to remember the ***direction*** of the field: imagine gripping the wire so that your ***right thumb*** points in the direction of the current; your fingers then curl in the direction of the lines of flux:

The diagrams show the magnetic field as you look down on the card:
Imagine the current direction as an arrow.
When the arrow moves away from you, into the page, you see the cross (×) of the tail of the arrow.
As the current flows towards you, you see the point of the arrow – the dot in the diagram.



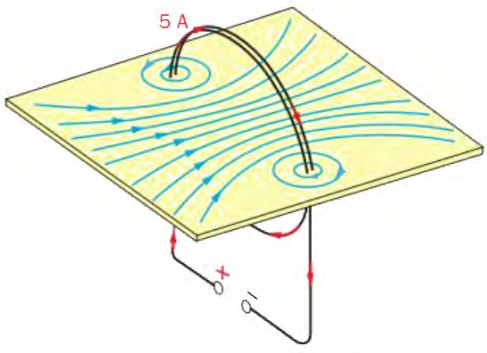


Can you see that the further from the wire the circles are, the more widely separated they become?
What does this tell you?

The flux density is greatest close to the wire.
As you move away from the wire the magnetic field becomes weaker.

The magnetic field of a flat coil

The diagram shows a flat coil carrying electric current:



Again, we can investigate the shape and direction of the magnetic field using iron filings and a compass.

Close to the wire, the lines of flux are circles.
Can you see that the lines of flux run anti-clockwise around the left side of the coil and clockwise around the right side?
What happens at the centre of the coil?

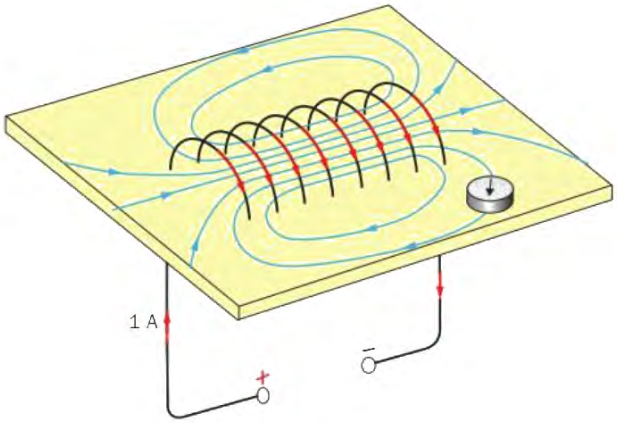
The fields due to the sides of the coil are in the same direction and they combine to give a strong magnetic field.

How would you expect the field to change if the direction of the current flow around the coil was reversed?

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The magnetic field of a solenoid

A **solenoid** is a long coil with a large number of turns of wire. Look at the shape of the field, revealed by the iron filings. Does it look familiar?

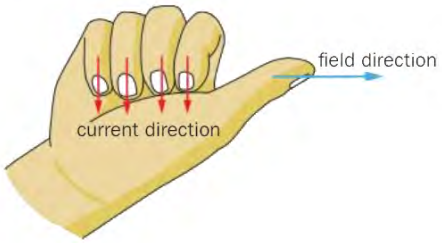


The magnetic field ***outside*** the solenoid has the ***same shape as the field around a bar magnet.***

Inside the solenoid the lines of flux are close together, parallel and equally spaced. What does this tell you?
For most of the length of the solenoid the flux density is constant. The field is uniform and strong.

If you reverse the direction of the current flow, will the direction of the magnetic field reverse?

A right-hand grip rule can again be used to remember the direction of the field, ***but*** this time you must curl the fingers of your ***right*** hand in the direction of the current as shown:



Your thumb now points along the direction of the lines of flux ***inside*** the coil – towards the end of the solenoid that behaves like the N-pole of the bar magnet.
This right-hand grip rule can also be used for the flat coil.

Magnetic materials

We have two different ways of producing magnetic fields: one uses permanent magnets, the other uses electric currents. Is there a link?

Electric current is the movement of charged particles (page 226). In 1864, James Clerk Maxwell proved that all magnetic fields are caused by the movement of charged particles.

Now think about the motion of the electrons in atoms.
As each electron spins, it acts like a tiny electric current and so it produces a very tiny magnetic field.
In some atoms the magnetic effects of all the electrons cancel; in others they do not, so that each atom acts like a tiny magnet.

In **ferromagnetic** materials, these tiny atomic magnets can line up with each other to produce a very strong magnetic field. Iron, cobalt and nickel are well-known ferromagnetic elements.



Using an electromagnet.

Why is the magnetic field of a solenoid greatly increased when a ferromagnetic core is placed inside the coil?
The atomic magnets of the core line up along the lines of flux inside the solenoid, and so the core becomes magnetised.

What happens when the current in the solenoid is turned off?
It depends on the material.

- A ***steel*** core stays magnetised. The tiny atomic magnets remain lined up, even when the external field is removed.
- An ***iron*** core quickly demagnetises, because the atomic magnets have enough vibrational energy to turn in random directions. A magnet that can be switched on and off is an **electromagnet**.



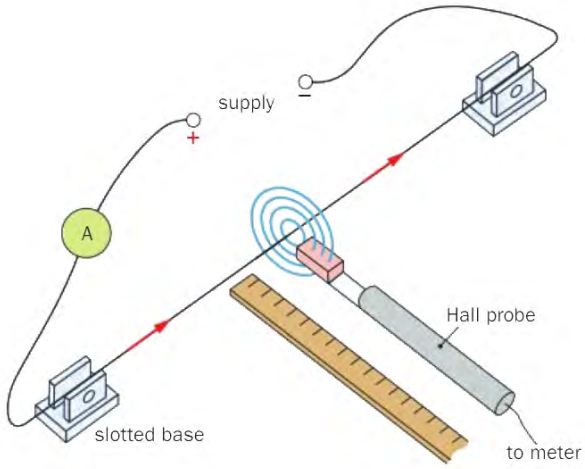
Removing a steel splinter from an eye.

Measuring magnetic flux density

We have seen the ***shapes*** of the magnetic fields around current-carrying conductors, but what do you think affects the ***strength*** or **magnetic flux density *B*** of these magnetic fields?

The long straight wire

You can investigate the magnetic flux density around a wire using the apparatus shown in the diagram:



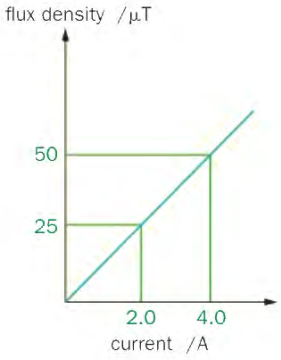
Remember that the lines of flux are circles all along the wire. How can you position the Hall probe so that the lines of flux pass at right angles through its tip?

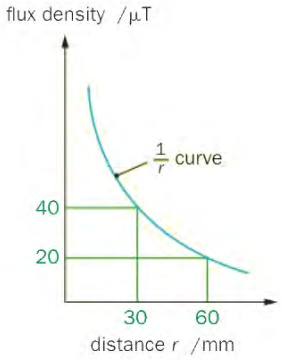
You can use the apparatus to answer the following questions:

1. When the probe is placed a fixed distance from the wire, how does *B* vary as the current in the wire is increased?
2. When the current is kept constant, how does *B* vary as the probe is moved away from the wire?

The graphs show some typical results. Can you see that:

- doubling the current ***doubles*** the flux density,
- doubling the distance from the wire ***halves*** the flux density.





In fact:

$$\text{Flux density, } B \text{ (tesla)} \propto \frac{\text{current, } I \text{ (amperes)}}{\text{distance, } r \text{ (metres)}} \quad \text{or} \quad B = k\frac{I}{r}$$

k is a constant. It depends on the material around the wire.

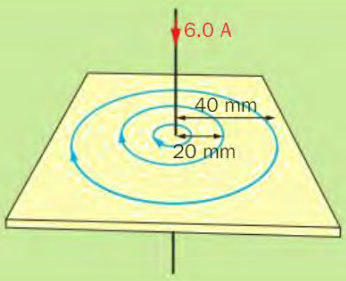
If the wire is in a vacuum, we usually write this as:

Magnetic flux density, $B = \dfrac{\mu_o I}{2\pi r}$

where μ_o is a constant called the ***permeability of free space***.
The permeability μ of a material is a measure of its effect on the strength of the magnetic field.
For a vacuum, $\mu_o = 4\pi \times 10^{-7}$ tesla metre ampere^{-1} (T m A^{-1}).

Our wire is in air, not a vacuum, and so we should really use the permeability of air (μ_{air}) in our equation, instead of μ_o.
In fact, air and most other materials have almost the same permeability as a vacuum.
But ferromagnetic materials have much higher permeabilities.
For example, the permeability of iron is about $8\pi \times 10^{-4}$ T m A^{-1}.

Example 1
A vertical wire carries a current of 6.0 A.
What is the flux density at (a) 20 mm and (b) 40 mm from the wire?



a) $B = \dfrac{\mu_o I}{2\pi r} = \dfrac{4\pi \times 10^{-7}\text{ T m A}^{-1} \times 6.0\text{ A}}{2\pi \times 20 \times 10^{-3}\text{ m}} = 6.0 \times 10^{-5}\text{ T}$

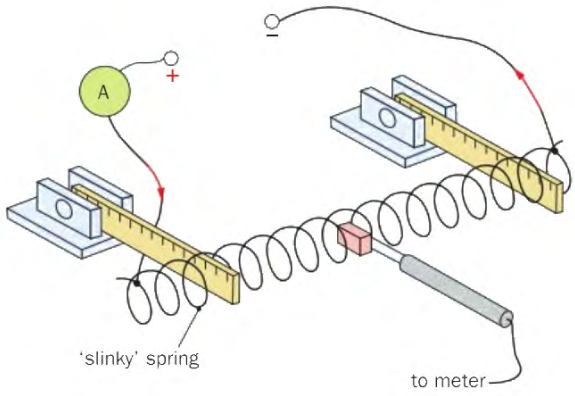
b) At twice the distance from the wire, the flux density is halved,

so: $B = \dfrac{6.0 \times 10^{-5}\text{ T}}{2} = 3.0 \times 10^{-5}\text{ T}$

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The solenoid

In the diagram, a 'slinky' is used as an adjustable solenoid:

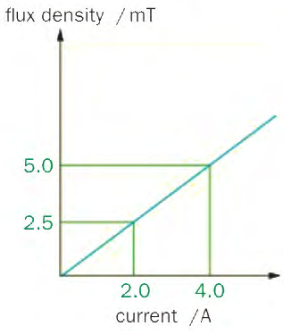


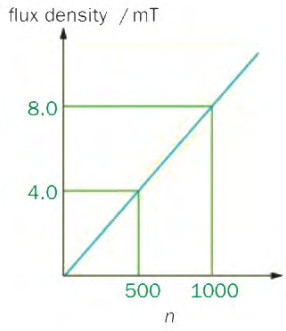
You can use this apparatus to answer the following questions:

1. Is B uniform at all points inside the solenoid?
2. How does the value of B at the centre of the solenoid vary as the current in the solenoid is increased?
3. How does the value of B within the solenoid vary as the solenoid is stretched (so that n, the number of turns per metre, decreases)?

The graphs show some typical results. Can you see that:

- doubling the current ***doubles*** the flux density,
- doubling the number of turns per metre ***doubles*** the flux density.





In fact, B is uniform inside the solenoid, and:

Magnetic flux density, B (tesla) $\propto$ turns per metre, n $\times$ current, I (amperes)

When the solenoid is in a vacuum (or in air), we can write this as an equation:

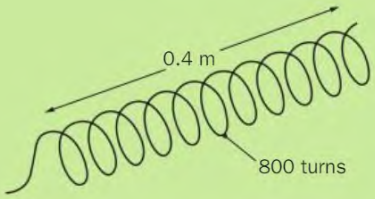
Magnetic flux density, $B = \mu_o\, n\, I$

Example 2

A solenoid is 0.40 m long and is made with 800 turns of wire.

a) What is n, the number of turns per metre?

b) What is the flux density at the centre of the solenoid when it carries a current of 2.0 A?



a) $n = \dfrac{\text{the number of turns}}{\text{the length of the solenoid}} = \dfrac{800}{0.40\ \text{m}} = 2000\ \text{m}^{-1}$

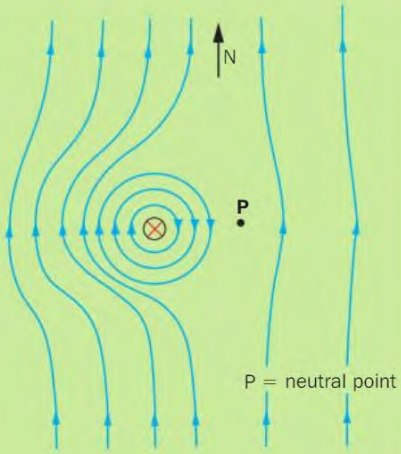
b) $B = \mu_o\, n\, I = 4\pi \times 10^{-7}\ \text{T m A}^{-1} \times 2000\ \text{m}^{-1} \times 2.0\ \text{A} = 5.0 \times 10^{-3}\ \text{T}$

Example 3

Phiz passes a current of 5.0 A down a vertical wire and uses a compass to plot the lines of flux around the wire. He obtains the pattern shown, and finds a neutral point P, 50 mm from the wire:

a) Why has he obtained this pattern?

b) What is the value of the horizontal component of the Earth's magnetic field?



a) This field is the ***resultant*** magnetic field of both the wire and the Earth in a horizontal plane.

b) At P, the two fields must be ***equal***, but in ***opposite*** directions.

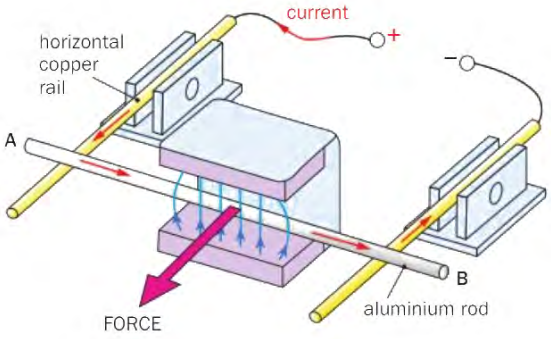
For the wire: $B = \dfrac{\mu_o I}{2\pi r}$

$$\therefore \quad B = \frac{4\pi \times 10^{-7}\ \text{T m A}^{-1} \times 5.0\ \text{A}}{2\pi \times 50 \times 10^{-3}\ \text{m}} = 2.0 \times 10^{-5}\ \text{T}$$

So the horizontal component of the Earth's field is also 2.0×10^{-5} T

Magnetic force

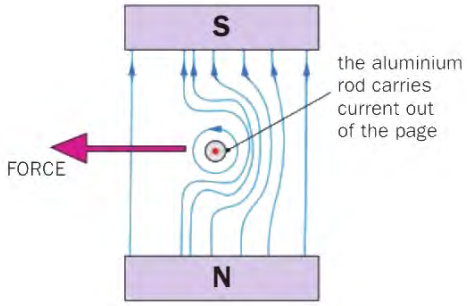
A wire carrying a current in a magnetic field feels a force.
A simple way to demonstrate this is shown in the diagram:



The two strong magnets are attached to an iron yoke with opposite poles facing each other. They produce a strong, almost uniform field in the space between them.

What happens when you switch the current on?
The aluminium rod AB feels a force, and moves along the copper rails as shown.
Notice that the current, the magnetic field and the force are ***all at right angles*** to each other.

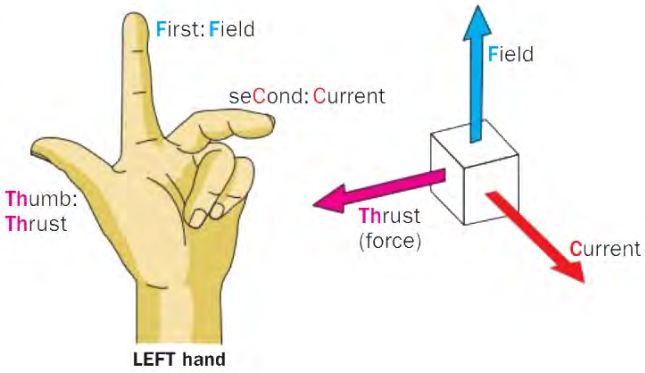
What happens if you reverse the direction of the current flow, ***or*** turn the magnets so that the magnetic field acts downwards?
In each case the rod moves in the opposite direction.



Why does the aluminium rod move?
The magnetic field of the permanent magnets interacts with the magnetic field of the current in the rod.
Imagine looking from end B of the rod. The diagram shows the combined field of the magnet and the rod:

The lines of flux behave a bit like elastic bands.
Can you see that the wire tends to be catapulted to the left?

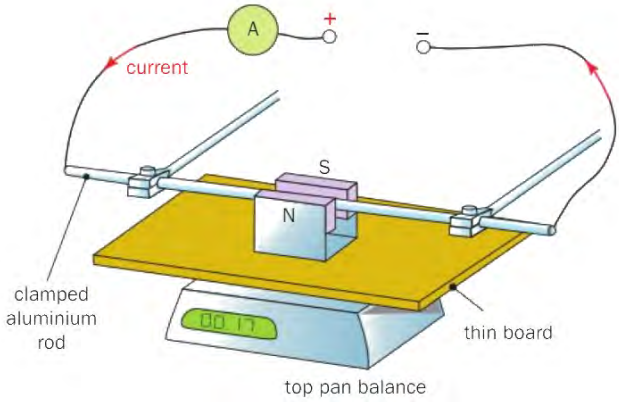
You can use **Fleming's left-hand rule** to predict the direction of the force.
You need to hold your ***left*** hand so that the thumb and the first two fingers are at right angles to each other as shown:



If your **F**irst finger points along the **F**ield direction (from N to S), and your se**C**ond finger is the **C**urrent direction (from + to −), then your **Th**umb gives the direction of the **Th**rust (or force).

What affects the size of the force?

Look carefully at the apparatus shown. The balance is set to read zero after the magnets have been placed on it.
A current is then passed along the clamped aluminium rod.



Can you use Fleming's left-hand rule to show that the force on the aluminium rod is acting upwards?
If the magnets exert an upward force on the rod, then, by Newton's Third Law, the rod must exert an equal but opposite force on the magnets. This downwards force will cause the reading on the balance to increase.

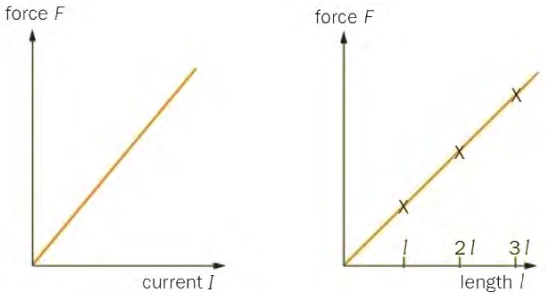
You can use the apparatus to answer the following questions:

1. How does the force vary when the current is increased?
2. How does the length of the rod in the field affect the force?
3. What happens to the force if the strength of the field is increased by adding extra magnets to the yoke?

Some typical results are shown on the next page.

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The graph of force against length was obtained by placing a second and a third pair of magnets on the balance. This doubles and then triples the length of the aluminium rod in the field. Look carefully at these graphs. What do they tell us?



Experiments like this show us that the force F on a conductor in a magnetic field is directly proportional to:

- the magnetic flux density B,
- the current I, and
- the length l of the conductor in the field.

In fact:

Force, F = flux density, B × current, I × length, l of wire
(newtons) (tesla) (amps) (metres)

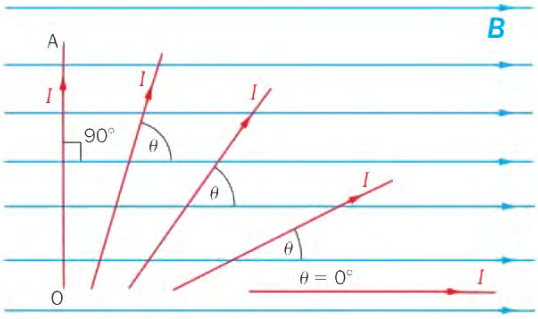
or **$F = B\,I\,l$**

This equation applies when the current is at 90° to the field. Does changing the angle affect the size of the force?

Look at the wire OA in the diagram, at different angles:

When the angle θ is 90° the force has its maximum value. As θ is reduced the force becomes smaller. When the wire is parallel to the field, so that θ is zero, the force is also zero.

In fact, if the current makes an angle θ to the magnetic field the force is given by:

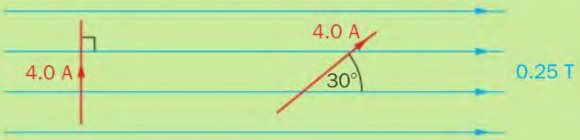


$F = B\,I\,l \sin\theta$

Notice that: when $\theta = 90°$, $\sin\theta = 1$ and $F = BIl$ as before.
when $\theta = 0°$, $\sin\theta = 0$ and $F = 0$, as stated above.

The ***size*** of the force depends on the angle that the wire makes with the magnetic field, but the ***direction*** of the force does not. The force is ***always at 90°*** to both the current and the field.

Example 4
The two wires below are in a uniform magnetic field of 0.25 T. Both wires are 0.50 m long and carry a current of 4.0 A. Calculate the size of the force on each wire.



a) The wire is at 90° to the field.
$F = B\,I\,l$
$\therefore\ F = 0.25\ \text{T} \times 4.0\ \text{A} \times 0.50\ \text{m} = \underline{0.50\ \text{N}}$

b) The wire is at 30° to the field.
$F = B\,I\,l \sin\theta$
$\therefore\ F = 0.25\ \text{T} \times 4.0\ \text{A} \times 0.50\ \text{m} \times \sin 30° = \underline{0.25\ \text{N}}$

In both cases the force acts at 90° to the wire – into the paper.

Magnetic flux density B and the tesla

We can rearrange the equation $F = BIl$ to give:

$$B = \frac{F}{Il}$$

What is the value of B, when $I = 1$ A and $l = 1$ m? In this case, B has the same numerical value as F.

This gives us the definition of B:
The magnetic flux density B is the force per unit length, acting on a wire carrying unit current, which is perpendicular to the magnetic field.

The unit of B is the **tesla** (T).
Can you see that: $1\ \text{T} = 1\ \text{N A}^{-1}\ \text{m}^{-1}$?

The tesla is defined in the following way:
A magnetic flux density of 1 T produces a force of 1 N on each metre of wire carrying a current of 1 A at 90° to the field.

The magnetic force on a moving charge

The photograph shows the glow of the 'Northern Lights'. Charged particles from outer space become trapped in the Earth's magnetic field and spiral from one pole to the other. As they enter the atmosphere at the poles, they produce this spectacular glow in the sky.



Aurora borealis: the 'Northern Lights'.

This phenomenon occurs because a charged particle feels a force when it moves through a magnetic field.
What factors do you think affect the size of this force?

The force F on the particle is directly proportional to:

- the magnetic flux density B,
- the charge on the particle Q, and
- the velocity v of the particle.

When the charged particle is moving at 90° to the field, the force can be calculated from:

Force, F (newtons) **= flux density, B** (tesla) **× charge, Q** (coulombs) **× velocity, v** ($m s^{-1}$)

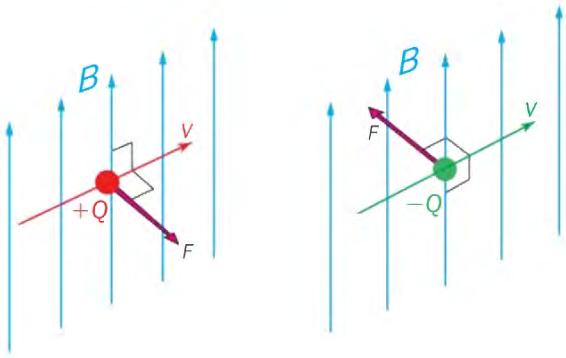
or

$$F = B Q v$$

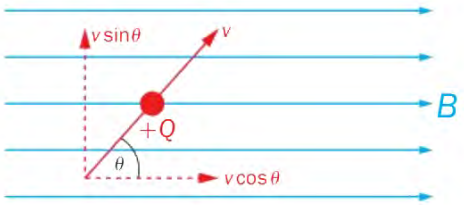
In which direction does the force act?
The force is always at 90° to both the current and the field, and you use ***Fleming's left-hand rule*** (page 262) to find its direction.

Try this for the positive particle in the first diagram:
Point your middle finger along the path of the positive particle.
Do you get the result shown?

How can you get the result for the negative particle?
(Note: the left-hand rule applies to conventional current flow.)
A negative charge moving to the right has to be treated as a positive charge moving to the left.
You must point your middle finger in the opposite direction to the movement of the negative charge.



What if the charged particle is moving at an angle θ to the magnetic field, with speed v?
We can resolve the velocity into two components as shown:
The component $v \sin\theta$ is at 90° to the magnetic field, and so the force on the particle is given by: $F = B Q v \sin\theta$



The component $v \cos\theta$ is not affected by the magnetic field, because it is parallel to it.

Example 5
An electron moving at 6.0×10^5 m s^{-1} passes perpendicularly through a magnetic field of 2.0×10^{-2} T.
The charge of the electron is 1.6×10^{-19} C.
What is the force on the electron?

$F = B Q v$
$F = 2.0 \times 10^{-2}\ \text{T} \times 1.6 \times 10^{-19}\ \text{C} \times 6.0 \times 10^5\ \text{m s}^{-1}$
$F = 1.9 \times 10^{-15}\ \text{N}$ (2 s.f.)

Do the equations $F = BIl$ and $F = BQv$ agree?
Suppose a charge Q moves a distance l, in t seconds.
Current is the rate of flow of charge, and so (from page 228): $I = \frac{Q}{t}$
Substituting for I in $F = BIl$, we get $F = \frac{BQl}{t}$.
But $\frac{l}{t}$ is the velocity v of the particle, so $F = BQv$.

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Circular paths

The movement of charged particles in magnetic fields is very important in atomic physics. In the 'fine beam tube' shown here, electrons move at right angles to a horizontal magnetic field. Can you see the path of the electron beam, a bright blue circle?

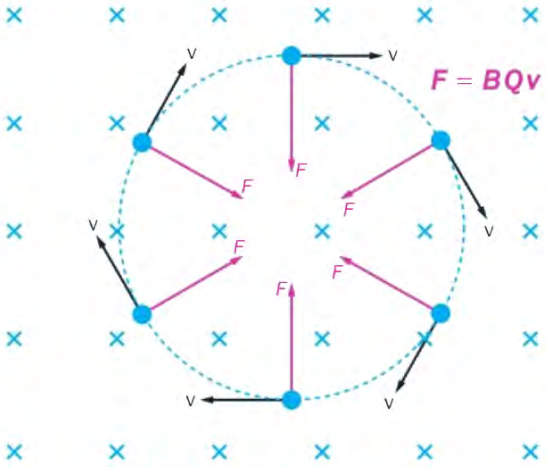


An electron beam in a fine beam tube.

What provide the ***centripetal force*** to make each electron move in a circular path? (see chapter 7)

Look at the diagram. It shows electrons moving in a uniform magnetic field. The flux density B is ***into*** the paper (as shown by the $\times$ symbols). Each electron in the beam is moving with a speed v, at 90° to the field.

The field exerts a force F on the electron, where $F = BQv$. Can you see that this force is always at 90° to the electron's velocity? Because of this, the magnetic force changes the direction, but not the speed, of the electron.
The BQv force provides the centripetal force.



The equation for centripetal force (see page 80) is:

$$F = \frac{mv^2}{r} \qquad \text{where } m = \text{the mass of the particle, and } r = \text{the radius of the circular path.}$$

The centripetal force is provided by the magnetic force, and so:

$$\frac{mv^2}{r} = BQv \qquad \text{or, rearranging:} \qquad \boldsymbol{r = \frac{m\,v}{B\,Q}}$$

Can you see from this that the electrons move in a larger circle if their speed is increased, or the magnetic field is made weaker?

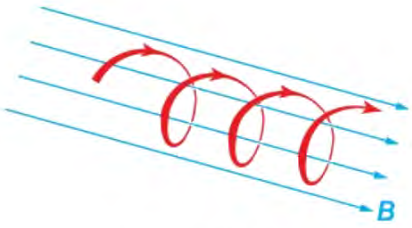
What is the time period T for the electron to make one rotation?

Since time $= \frac{\text{distance}}{\text{speed}}$, $T = \frac{\text{length of circular path}}{\text{speed of electron}}$ or $T = \frac{2\pi r}{v}$

but since $r = \frac{mv}{BQ}$, we get: $T = \frac{2\pi}{v} \times \frac{mv}{BQ}$ so: $\boldsymbol{T = \frac{2\pi\,m}{B\,Q}}$

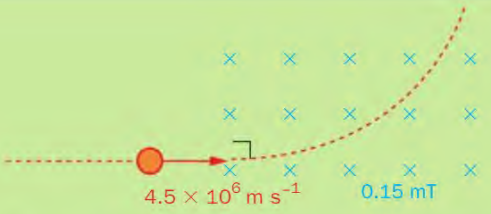
In a field of constant flux density B, the time period T of the electron does not depend on its speed. A faster electron moves in a circle of larger radius, but it takes exactly the ***same time*** for one revolution.

What happens when the particle is not moving at 90° to the field? The $\boldsymbol{v \sin\theta}$ component makes it move in a circle. The $\boldsymbol{v \cos\theta}$ component makes it move parallel to the lines of flux (see page 260). This means that the particle spirals around the field lines, as shown:



Spiralling particle.

Example 6
A particle of mass 9.1×10^{-31} kg and of charge 1.6×10^{-19} C is moving at 4.5×10^6 m s^{-1}. It enters a uniform magnetic field of flux density 0.15 mT, at 90°, as shown in the diagram:
What is the radius of its circular path?



Using: $r = \frac{mv}{BQ}$ (as derived above) $\qquad 1\text{ mT} = 10^{-3}\text{ T}$

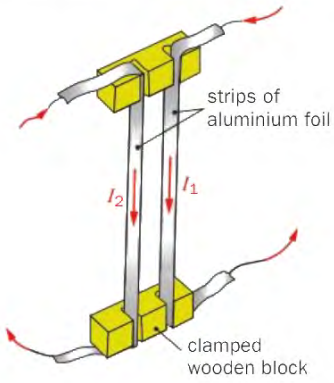
$$\therefore r = \frac{9.1 \times 10^{-31}\text{ kg} \times 4.5 \times 10^6\text{ m s}^{-1}}{0.15 \times 10^{-3}\text{ T} \times 1.6 \times 10^{-19}\text{ C}} = \underline{0.17\text{ m}}$$

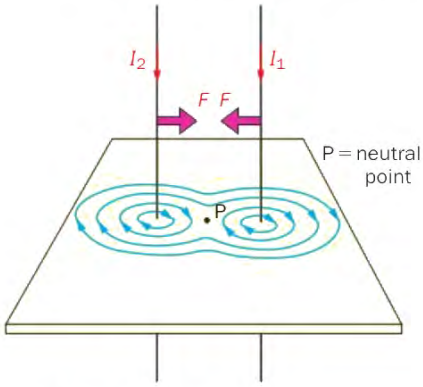
You can use the left-hand rule to show that the particle must be ***positively*** charged.

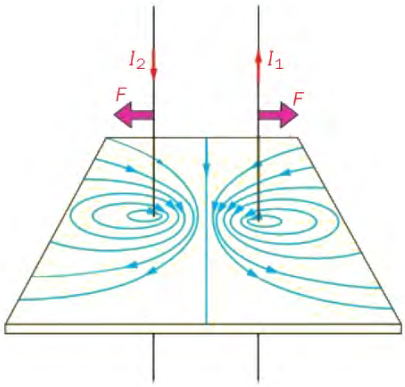
Forces between currents

What happens when current is passed along two strips of foil as shown below?
The strips bend as they attract or repel each other.

Two parallel, current-carrying wires exert equal but opposite forces on each other.
Look carefully at these forces, and the resultant magnetic fields around the wires.

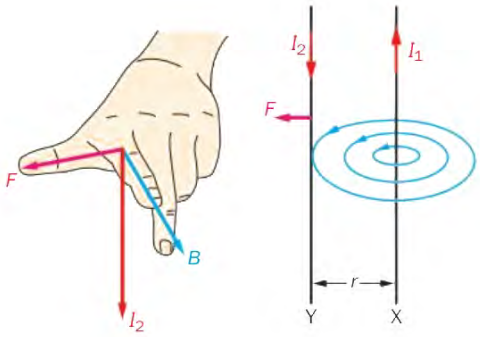






Notice that: currents flowing in the ***same*** direction ***attract***,
currents flowing in ***opposite*** directions ***repel***.

How do these forces arise?
The diagram shows the anti-clockwise field around wire X:
Wire Y is at 90° to this field, and so it experiences a force.
Apply Fleming's left-hand rule (page 262) to wire Y.
Do you find that the force on wire Y is to the left, as shown?



What is the size of the force?
Notice that the wires are a distance r apart.
Wire X carries a current I_1. Wire Y carries a current I_2.
What is the flux density B at wire Y, due to the current I_1 in X?
From page 260 we know that:

$$B = \frac{\mu_o I_1}{2\pi r} \qquad \ldots (1)$$

What is the force F on a length l of wire Y?
From page 263 we know that:

$$F = BI_2 l \qquad \ldots (2)$$

If we use equation (1) to replace B in equation (2):

$$F = \frac{\mu_o I_1 I_2 l}{2\pi r} \qquad \ldots (3)$$

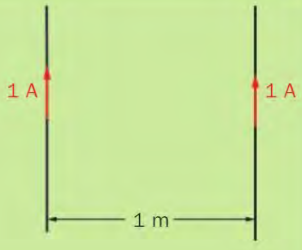
Similarly, wire X feels a force due to the field of wire Y.
Can you show that equation (3) also gives the force on wire X?
Use the left-hand rule to prove that the force on X is to the right.

The definition of the ampere

The unit of current, the ampere, is defined in terms of the force between two currents.

As shown in Example 7 below, when two long wires are parallel and placed 1 metre apart in air, and if the current in each wire is 1 ampere, then the force on each metre of wire is 2×10^{-7} N.

Example 7
Two long parallel wires are placed 1.0 m apart in air as shown.
Both wires carry a current of 1.0 A.
What is the force acting per unit length on each wire?



$F = \frac{\mu_o I_1 I_2 l}{2\pi r}$ where $l = 1.0$ m and $\mu_o = 4\pi \times 10^{-7}$ T m A^{-1}

$$F = \frac{4\pi \times 10^{-7}\ \text{T m A}^{-1} \times 1.0\text{A} \times 1.0\text{A} \times 1.0\ \text{m}}{2\pi \times 1.0\ \text{m}} = 2 \times 10^{-7}\ \text{N}$$ on each metre

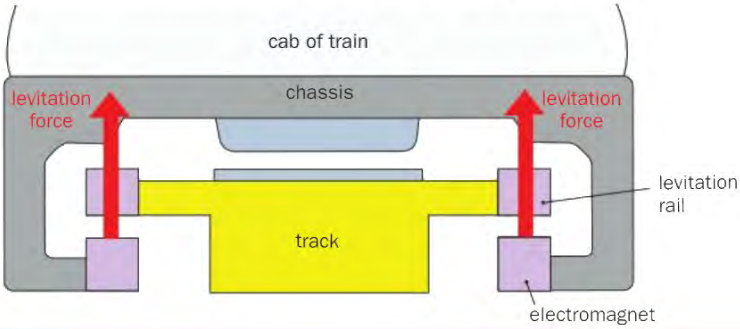
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Physics at work: Magnetic fields

The magnetic levitation train

The 'maglev' train in the photograph does not need wheels, because it 'floats' above its track by ***magnetic levitation***. The diagram below shows how magnetic levitation works.

The levitation magnets are powerful electromagnets. Can you see that they are attached to the chassis of the train? The magnets are ***attracted*** upwards towards the levitation rail. This force lifts the train 15 mm above its T-shaped track.





How does the train maintain its exact position above the track? Computers control the current in the electromagnets, and so the strength of the levitation force can be varied.

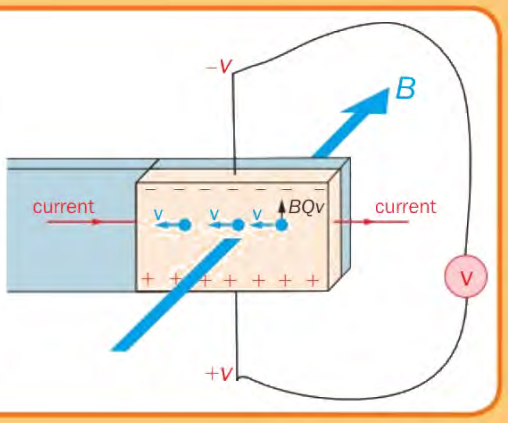
The Hall effect

How does a Hall probe measure the strength of a magnetic field? The diagram shows the semiconducting slice at the tip of a Hall probe. A small current is passed through the slice as shown: The electrons are moving with a velocity v.

When a magnetic field is applied, the BQv force pushes these electrons towards the upper edge of the slice. This becomes negatively charged. The lower edge, which is short of electrons, becomes positively charged.

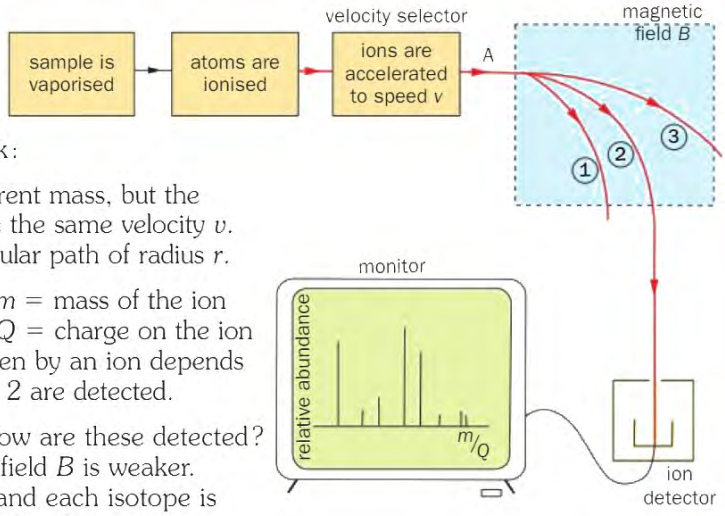
If a voltmeter is placed across the top and bottom edges of the slice, it detects a small p.d. The stronger the field, the larger the p.d.

You can calibrate the voltmeter so that it reads in tesla, instead of volts.



The mass spectrometer

Mass spectrometers are used to identify the different ***isotopes*** (see page 357) in a sample of material, and measure their relative abundance. The diagram shows the principle of how they work:



Each isotope in the sample forms ions with a different mass, but the velocity selector ensures that all the ions at A have the same velocity v. The magnetic field B deflects these ions into a circular path of radius r.

From page 265, r is given by: $r = \dfrac{mv}{BQ}$ where m = mass of the ion and Q = charge on the ion

Since B and v are fixed, the radius of the path taken by an ion depends on its value of m/Q. Only the ions following path 2 are detected.

Ions with a smaller value of m/Q take path 1. How are these detected? All the ions are deflected less when the magnetic field B is weaker. The applied magnetic field is gradually increased and each isotope is detected in turn. The screen shows their relative abundance.

Summary

Permanent magnets and electric currents produce magnetic fields. We show fields by drawing lines of flux. The field round a straight wire is in concentric circles. The field inside a solenoid is strong and uniform.

Magnetic flux density B is a vector quantity and is measured in tesla (T). $1\ \text{T} = 1\ \text{N A}^{-1}\ \text{m}^{-1}$

Around a long straight wire, $B = \dfrac{\mu_0 I}{2\pi r}$ Inside a solenoid, $B = \mu_0 n I$

μ_0 is called the permeability of free space (a vacuum).

The force on a current in a wire at an angle θ to a magnetic field is: $F = B I l \sin\theta$

The force on a charge moving at an angle θ to a magnetic field is: $F = B Q v \sin\theta$; when $\theta = 90°$, $F = B Q v$

The direction of the force on the current, and on the moving charged particle, is given by Fleming's ***left***-hand rule.

There is a force between two parallel current-carrying wires: this force is used to define the ampere.

Questions

1. Can you complete these sentences?

a) A magnetic ..... can be represented by lines of flux. The more closely packed the lines of flux, the ..... the magnetic field.

b) There is always a ..... field around a wire carrying an electric .....
When the wire is straight and long, the lines of flux are ..... around the .....

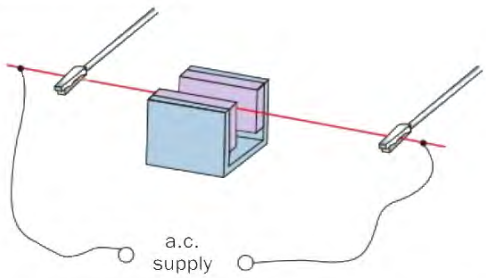
c) ***Inside*** a solenoid, the lines of flux are ..... and equally ..... This shows that the field is .....
Outside, the field is like that of a ..... magnet.

d) When a current crosses a magnetic field, it experiences a ..... The direction of this ..... is at ..... to both the current and the ..... directions.

e) The magnetic flux density of a solenoid can be increased by increasing the number of ..... per metre, and increasing the .....

2. Explain the following:

a) A wire clamped firmly at both ends and placed between the poles of a horseshoe magnet, as shown below, vibrates when alternating current is passed through it:



b) A charged particle moving at 90° to a uniform magnetic field always moves in a circular path.

c) Two long, parallel wires carrying electric current in opposite directions repel each other.

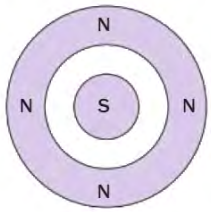
3. Sketch the magnetic field patterns:

a) between the poles of a horseshoe magnet,

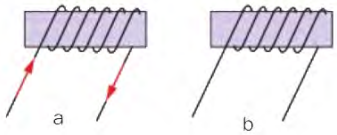
b) around a long, straight wire carrying electric current vertically upwards,

c) around two long parallel wires carrying current in the same direction,

d) between the poles of the cylindrical magnet below.



4. The diagram below shows a pair of iron-cored solenoids. The solenoids attract each other. Copy the diagram and add the direction of the current flow in solenoid b.



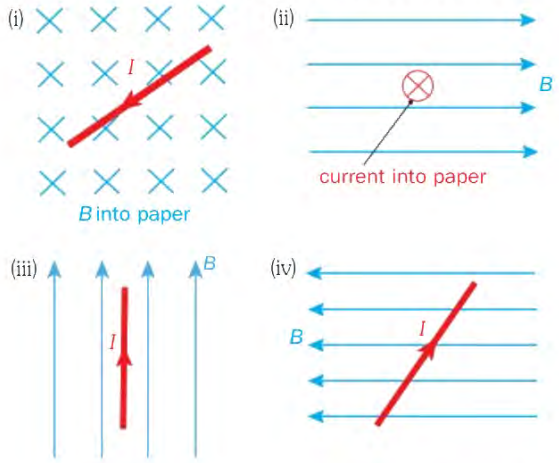
5. Calculate the magnetic flux density at a distance of 5.0 cm from a long, straight wire carrying a current of 4.0 A.

6. The magnetic flux density at a point close to a long wire carrying a current of 2.0 A is 2.0×10^{-5} T. How far is this point from the wire?

7. A solenoid of length 40 cm has 1200 turns. What is the magnetic flux density at the centre of the solenoid when the current through it is 2.5 A?

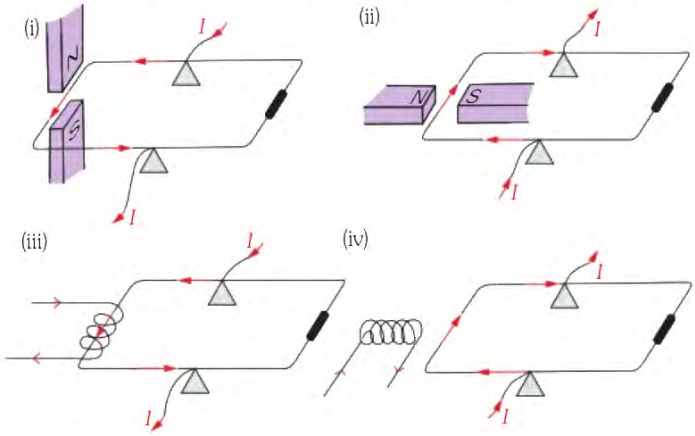
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8. The diagrams below show 4 examples of conductors in magnetic fields.
a) In which of the diagrams will there be a force acting on the conductor?
b) If there is a force, in which direction does it act?

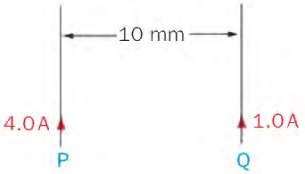


9. A solenoid can carry a maximum current of 10 A. If it is to produce a field of flux density of 0.20 T, how many turns per metre would the solenoid need?

10. Each diagram below shows a ***current balance***. This is a rectangular piece of wire supported on pivots so it balances. Which of these current balances will tilt when current flows through the balance as shown?

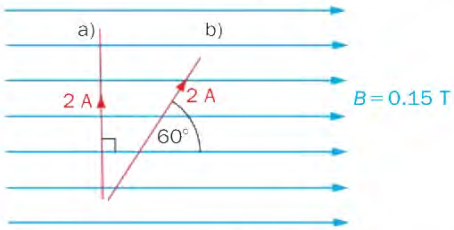


11. Two long, parallel, vertical wires P and Q are 10 mm apart. Both wires carry current upwards. The current in P is 4.0 A; the current in Q is 1.0 A.



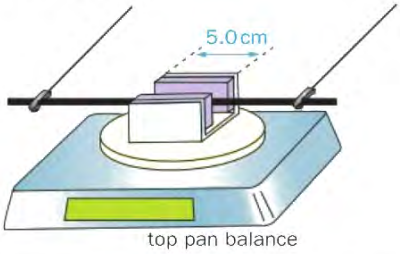
a) What is the force on each metre length of wire?
b) There is a neutral point between the two wires. How far is this neutral point from wire P?

12. A wire of length 0.40 m is placed in a field of flux density 0.15 T as shown below.
The current in the wire is 2.0 A.
Calculate the size of the force exerted on the wire:
a) when the wire is at 90° to the magnetic field,
b) when the wire is at an angle of 60° to the field.

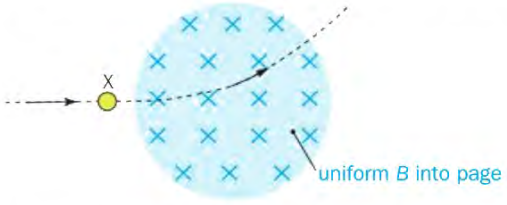


c) What is the direction of the force in each case?

13. When a current of 4.0 A is passed along the clamped wire shown in the diagram below, the reading on the balance increases by 2.2 g. What is the flux density of the field between the poles of the magnet if the length of the wire in the field is 5.0 cm?
(Take g to be 10 N kg^{-1}.)



14. A particle X enters a uniform magnetic field.
a) Is X positively or negatively charged?
b) What will happen to the path of particle X if:
i) it is slowed down,
ii) the strength of the magnetic field is increased,
iii) the direction of the magnetic field is reversed?



15. An electron moving at 2.0×10^7 ms^{-1} passes at 90° through a magnetic field of flux density 2.5×10^{-2} T. The electron has a mass of 9.1×10^{-31} kg and a charge of -1.6×10^{-19} C.
a) What force is exerted on the electron by the field?
b) What is the radius of the electron's circular path in the field?
c) What happens to the path taken by the electron if it enters at 30° to the magnetic field instead of 90°?

Further questions on page 315.

19 Electromagnetic Induction

The photograph shows a wind farm. Each aerogenerator converts the kinetic energy of the wind into electrical energy. However, its electricity output is small when compared to the 50 gigajoules of energy that Britain's power stations can produce in each second.



How is all this electrical energy produced? Power stations generate electricity by using ***electromagnetic induction***. This was discovered in 1831 by Michael Faraday.

In this chapter you will learn:

- about magnetic flux and magnetic flux linkage,
- how to use Faraday's Law and Lenz's Law,
- some of the uses of electromagnetic induction.

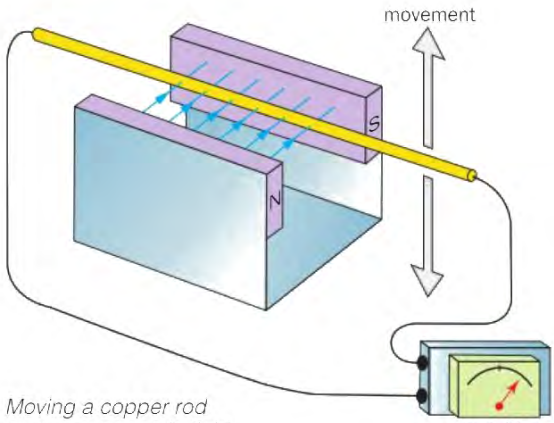
What is electromagnetic induction?

The diagram shows a copper rod connected to an ammeter: There is no battery in the circuit.

What happens when you move the copper rod downwards, to cut ***across*** the horizontal magnetic field?
The pointer on the meter makes a brief 'flick' to the right, showing that an electric current has been **induced**.

What happens when you move the rod upwards?
The meter again gives a 'flick', but this time to the left.
You have now induced a current in the opposite direction.

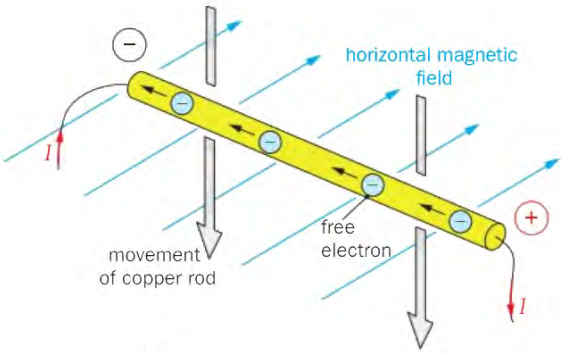
If you hold the rod stationary, or if you move the rod ***along*** the field lines, there is no induced current.



Moving a copper rod across a magnetic field.

Why does electromagnetic induction occur?

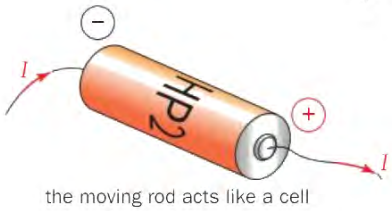
When you move the copper rod, its free electrons move with it. But when a charge moves in a magnetic field it experiences a force on it (the BQv force, see page 264).
You can use Fleming's left-hand rule (page 262) to show that the force on each electron is to the left, as shown in the diagram: (Remember – an electron moving down has to be treated as a positive charge moving up.)



So electrons accumulate at one end of the rod, making it negative. This leaves the other end short of electrons and therefore positive. There is now a ***voltage*** across the ends of the moving rod.

If the ends of the moving rod are joined to form a complete circuit, the induced voltage causes a current to flow round the circuit, as shown by the 'flick' of the ammeter.
The induced voltage is a source of electrical energy – an ***e.m.f.***

When a conductor is moving in a magnetic field like this, ***an e.m.f. is induced***, even if there isn't a complete circuit for a current to flow.



the moving rod acts like a cell

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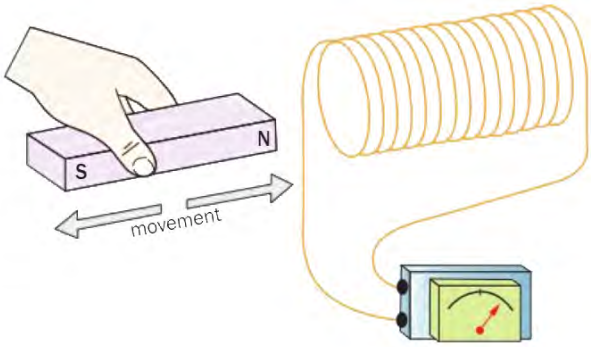
Faraday's model of electromagnetic induction

The diagram shows another way to demonstrate induction:

What happens when you move the magnet ***into*** the coil?
An e.m.f. is induced across the ends of the coil. This e.m.f. drives a current round the circuit and the meter gives a 'flick'.

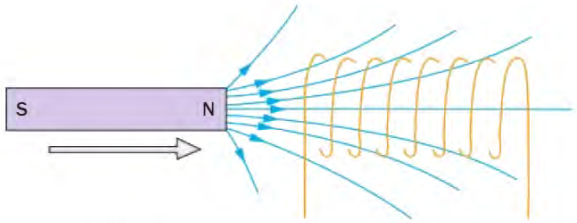
If you move the magnet ***out of*** the coil, then an e.m.f. is induced in the reverse direction, and the ammeter 'flicks' the opposite way.

What happens if you move the coil instead of the magnet?
The meter still shows a current. An e.m.f. is induced whenever there is ***relative movement*** between the magnet and the coil.



Michael Faraday devised a model to help us visualise what is happening in electromagnetic induction.
Think about the experiment on the facing page.
As the rod moves downwards, can you visualise it 'cutting' through the lines of flux between the N-pole and the S-pole?

In Faraday's model, this cutting of the lines of magnetic flux causes an e.m.f. to be induced across the conductor.

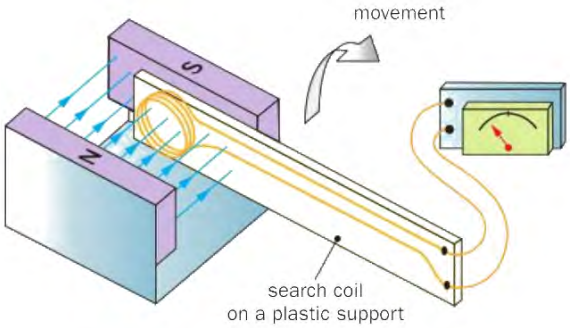


The coil cuts through the lines of flux.

Does this idea also work for the magnet and coil?
Look at the diagram. As the magnet is moved towards the coil, each turn of the coil cuts the lines of flux of the bar magnet.

The more turns on the coil, the greater the e.m.f. This is because N turns will cut N times more lines of flux than just one turn.

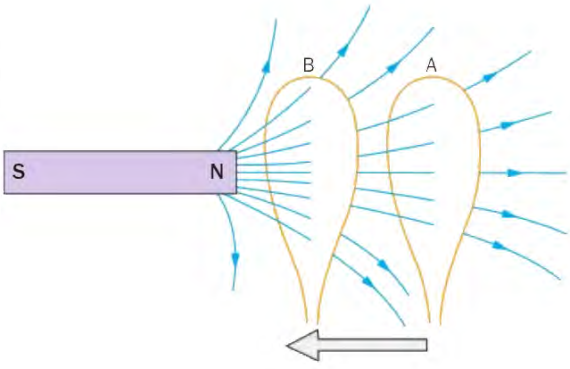
Faraday's flux model can be looked at in a different way.
The diagram shows a small flat coil, called a search coil, connected to an ammeter:
If the coil is pulled quickly out of the field, the meter gives a flick.
In the diagram, can you see three lines of flux passing through or ***linking*** the coil, when it is in the field?
When it is out of the field the flux linkage of the coil is zero.
This ***change in flux linkage*** induces an e.m.f. across the coil.



So Faraday's model tells us that an e.m.f. is induced when
- a conductor like the copper rod cuts through lines of flux, or
- when there is a change in the flux linkage of a coil.

Flux cutting and changes in flux linkage are just two ways of looking at the same thing. Look at the lines on the diagram:

When the coil is moved from A to B we can say that the number of lines of flux linking it changes from 5 to 9 – a change of **4**.
Or we can say that the coil has cut through **4** lines of flux.
These are two ways of describing the same change.



You can increase the size of the e.m.f. induced across a coil by:
- using a stronger magnet,
- using a coil with more turns of wire,
- using a coil with a greater cross-sectional area,
- moving the magnet faster.

This is because the larger and the faster the change in flux linkage of the coil, the greater the induced e.m.f.

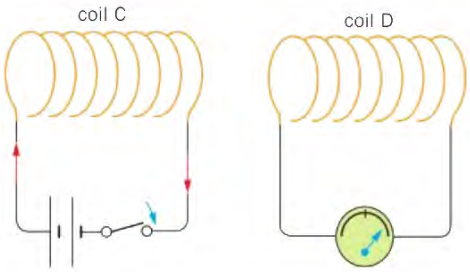
Electromagnetic induction using two coils



Michael Faraday.

Faraday was not using a magnet when he first discovered electromagnetic induction. He was using two coils, wrapped around an iron ring. What did he see?

The diagram shows two coils placed side by side:
Coil C is connected to a battery and a switch.
Coil D is connected to a sensitive ammeter.
Think about these 4 questions:

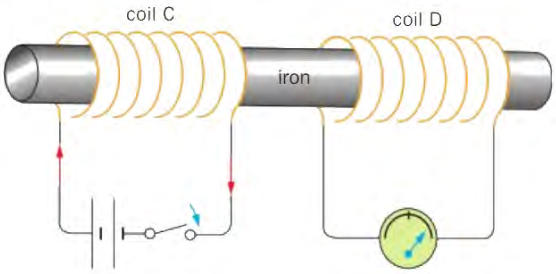


1. What do you think happens when the switch is closed?
The meter gives a flick to the right, and then goes back to zero. (You would see the same effect if you were to move the N-pole of a bar magnet quickly into coil D.)

2. Why is a current induced in coil D?
When you switch on the current, a magnetic field 'grows' around coil C. The 'growing' flux lines link the turns of coil D. As its ***flux linkage changes***, an e.m.f. is induced across coil D and this induced e.m.f. drives a current through the ammeter.

3. Why does the ammeter reading go back to zero?
When the current in coil C reaches its final steady value, the magnetic field around C is steady, so the flux linkage of D is ***not changing***, and so there is no induced e.m.f. and no current.

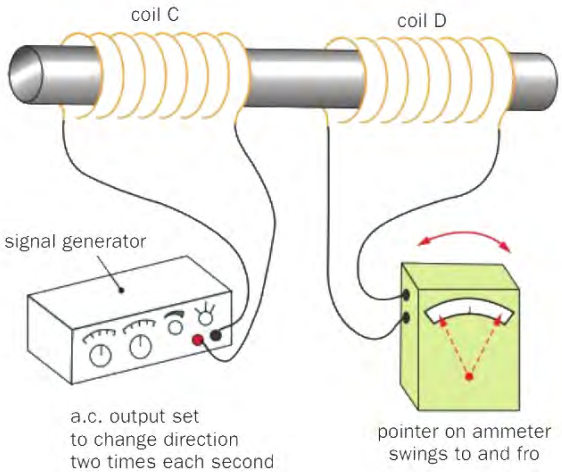
4. What do you think will happen when the switch is opened?
As the current in C falls, its magnetic field collapses.
As the flux linkage of D ***changes***, an e.m.f. is induced again.
The meter gives a flick again, but this time to the left, because the induced e.m.f. and the current are in the opposite direction.



Can you explain why the e.m.f. induced across coil D can be increased by:

- using an iron core, as shown in this diagram?
- using coils with a greater number of turns?
- passing a larger current around coil C?
- making the current in C change more quickly?

Remember: the larger and the faster the change in flux linkage of coil D, the greater the e.m.f. induced across it.



If the current in coil C is switched on and off repeatedly, what happens in coil D?
You would see the pointer on the meter move to and fro as the current in coil D constantly changes direction.
This is alternating current or a.c.

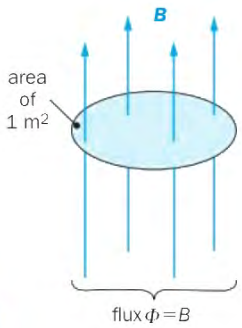
Now suppose you pass alternating current through coil C.
The field around C will grow and decay in one direction, and then grow and decay in the opposite direction.
The magnetic flux linkage of coil D will always be changing.
Passing a.c. through coil C induces a.c. in coil D, as shown:

This apparatus is a simple **transformer** (see also page 284).
Coil C is called the ***primary*** coil; coil D is the ***secondary*** coil.

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Magnetic flux, Φ

We draw lines of magnetic flux so that the density of the lines shows the strength of the field. So the number of lines passing at 90° through unit area represents the magnetic flux density B (see page 257).

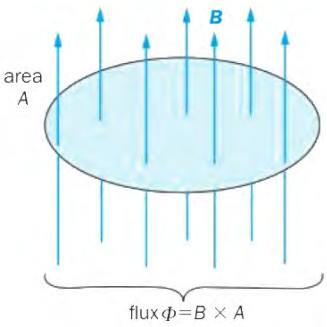


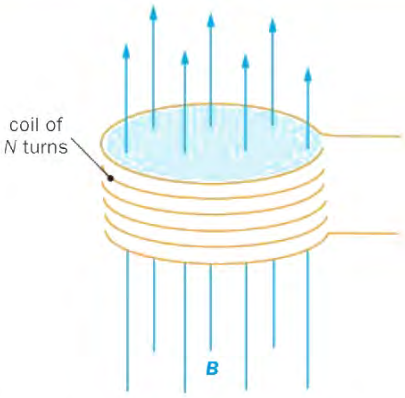
Magnetic flux has the symbol Φ (the Greek letter phi). It represents the ***total number*** of lines of flux that 'link', or pass at 90° through, a given area. Magnetic flux is measured in webers (Wb).

If B is the flux per unit area, the total flux linking the area A can be calculated from:

Magnetic flux, Φ = magnetic flux density, B × area, A
(weber) (tesla) (m^2)

or **$\Phi = BA$**

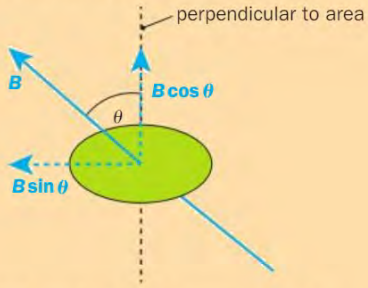




What is the flux linkage of the ***coil*** of cross-sectional area A and N turns? Each ***turn*** of the coil has flux Φ linking it, where $\Phi = BA$, and so:

Total flux linkage of the coil = $N\Phi = NBA$

Here, the perpendicular to the area A makes an angle θ with the flux density B. We can resolve B into two components:



$B\sin\theta$ ***does not link*** the area. The flux linkage is only by the $B\cos\theta$ component. So $\Phi = B\cos\theta \times A$, or:

Magnetic flux, $\Phi = BA\cos\theta$

A ***change*** in flux linkage is given the symbol $\Delta\Phi$. The flux linking a coil may change for two reasons: either (1) the flux density B changes, or (2) the area A linked by the flux changes.

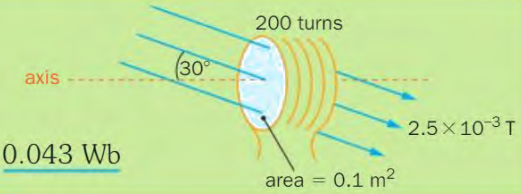
(1) If a single turn coil of area A is placed at 90° to a magnetic flux density that is changing, then:

$\Delta\Phi = \Delta B\, A$ where ΔB is the ***change in flux density***

(2) If a single turn coil moves in a flux density B so that the area linked by the flux changes, then:

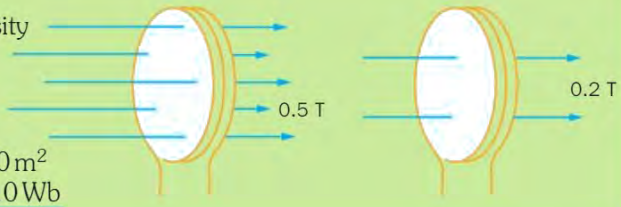
$\Delta\Phi = B\,\Delta A$ where ΔA is the ***change in area***

Example 1
How much flux is linking a 200 turn coil of area 0.10 m^2, when it is placed at 30° to a magnetic field $B = 2.5 \times 10^{-3}$ T?



Total flux linkage $= N\Phi = NBA\cos\theta$
$= 200 \times 2.5 \times 10^{-3}\,\text{T} \times 0.10\,\text{m}^2 \times \cos 30° = \underline{0.043\,\text{Wb}}$

Example 2
A coil of 100 turns and area 0.20 m^2 is at 90° to a flux density which decreases from 0.50 T to 0.20 T.
What is the change in flux linkage of the coil?



Change in flux linkage $= N\Delta\Phi = NB\Delta A$
$= 100 \times (0.50 - 0.20)\,\text{T} \times 0.20\,\text{m}^2$
$= 100 \times 0.30\,\text{T} \times 0.2\,\text{m}^2 = \underline{6.0\,\text{Wb}}$

Faraday's Law

We know that an e.m.f. is induced when there is a change in the flux linking a conductor.

Think again about the experiments on pages 270–272.
Do you agree that they all show that the larger and the faster the change in flux linkage, the greater the induced e.m.f.?



Phiz makes the link with Faraday's law.

Faraday's Law makes the connection between the size of the induced e.m.f. and the ***rate*** at which the flux is changing. It states that:

the magnitude of the induced e.m.f. is equal to the rate of change of flux linkage.

If a change in flux $\Delta\Phi$ occurs in a time Δt, then:

Induced e.m.f., ϵ (volts) $= \dfrac{\textbf{change in flux linkage, } \Delta\Phi \text{ (weber)}}{\textbf{time taken, } \Delta t \text{ (s)}}$

or $\epsilon = \dfrac{\Delta\Phi}{\Delta t}$

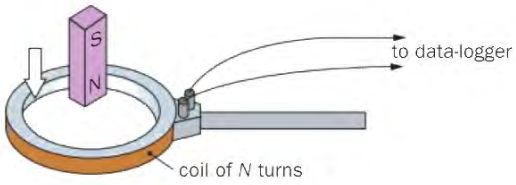
For a coil with N turns, the e.m.f. will be N times as great, and so:

$\epsilon = N\dfrac{\Delta\Phi}{\Delta t}$

Remember (from page 273): $\Delta\Phi$ can be calculated using
either $\Delta\Phi = \Delta BA$ ***or*** $\Delta\Phi = B\Delta A$

Dropping a bar magnet

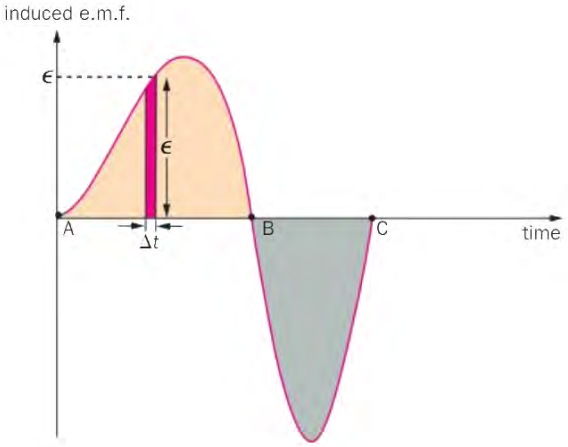
The diagram shows a bar magnet falling through a horizontal coil:



The coil is connected to a data-logger, which records the induced e.m.f. every millisecond.
The graph below shows how the induced e.m.f. varies with time:

Section AB shows us that as the magnet enters the coil, the flux linkage of the coil increases and an e.m.f. is induced.

Why is the induced e.m.f. negative over section BC?
As the magnet leaves the coil, the flux linkage decreases and the e.m.f. is induced in the opposite direction.



Rearranging Faraday's equation we get: $\epsilon\Delta t = N\Delta\Phi$

So $\epsilon\Delta t$ is equal to the flux change of the coil in the time Δt.

Can you see the area $\epsilon\Delta t$: the dark pink strip on the graph?
This small area tells us the change in flux linkage over time Δt.

Therefore, the total area under the graph for part AB (shaded pink) is the total change in flux linkage as the magnet enters the coil.

The area under the graph for the section AB (pink) is equal to the area under the graph for the section BC (shaded grey).
What does this tell us about the total change in flux linkage as the magnet leaves the coil and as the magnet enters the coil?

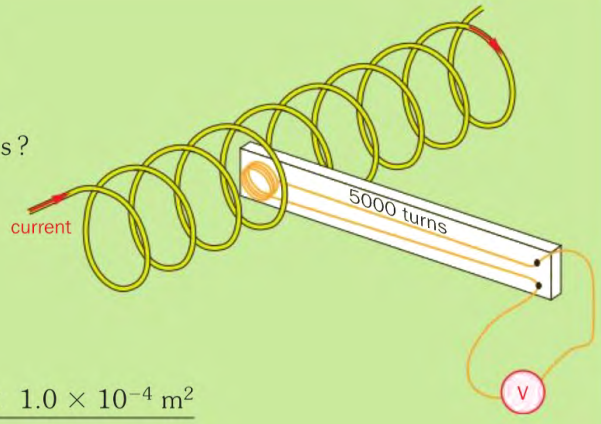
Why is the peak BC narrower but taller than the peak AB?
Remember the magnet is accelerating as it falls.

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Solving problems using Faraday's Law

Example 3

A search coil has 5000 turns and an area of 1.0×10^{-4} m^2. It is placed in a long current-carrying coil as shown, so that its face is at 90° to the lines of magnetic flux inside the coil. What e.m.f is induced across the coil when the flux density inside it changes from 2.5×10^{-3} T to 1.3×10^{-3} T in 0.40 s?



Note that it is the ***flux density*** linking the coil that changes.

$$\textbf{e.m.f.,}\ \epsilon = N\frac{\Delta\Phi}{\Delta t} \qquad \text{where} \quad \Delta\Phi = \Delta B\, A$$

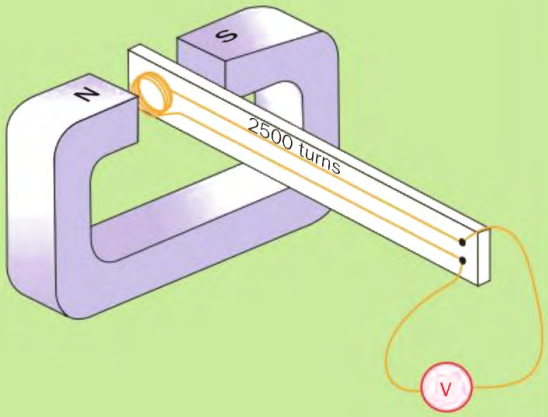
$$= N\frac{\Delta BA}{\Delta t}$$

$$= \frac{5000 \times (2.5 \times 10^{-3} - 1.3 \times 10^{-3})\ \text{T} \times 1.0 \times 10^{-4}\ \text{m}^2}{0.40\ \text{s}}$$

$$= \underline{1.5 \times 10^{-3}\ \text{V}} \quad (2\ \text{s.f.})$$

Example 4

A search coil has 2500 turns and an area of 1.5×10^{-4} m^2. It is placed between the poles of a horseshoe magnet, as shown. As the coil is rapidly pulled out of the field in 0.30 seconds, the average value of the e.m.f. induced across the coil is 0.75 V. Calculate the flux density between the poles of the magnet.



Note that as the coil moves, the ***area*** linked by the flux changes.

$$\textbf{e.m.f.,}\ \epsilon = N\frac{\Delta\Phi}{\Delta t} \qquad \text{where} \quad \Delta\Phi = B\,\Delta A$$

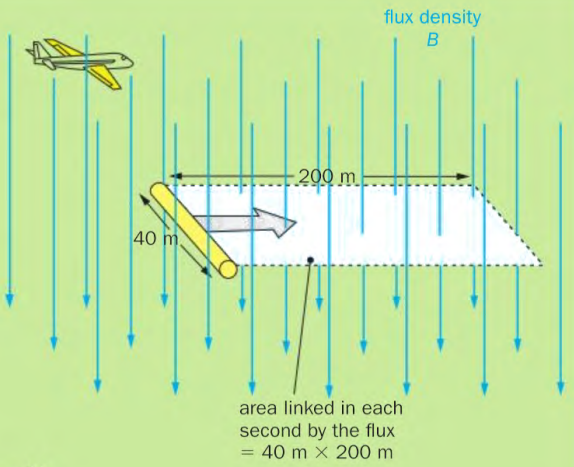
$$= N\frac{B\Delta A}{\Delta t}$$

$$0.75\ \text{V} = \frac{2500 \times B \times (1.5 \times 10^{-4} - 0)\ \text{m}^2}{0.30\ \text{s}}$$

$$\therefore\ B = \frac{0.75\ \text{V} \times 0.30\ \text{s}}{2500 \times 1.5 \times 10^{-4}\ \text{m}^2} = \underline{0.60\ \text{T}} \quad (2\ \text{s.f.})$$

Example 5

An aeroplane with a wingspan of 40 m moves at 200 m s^{-1}. It flies horizontally through an area where the vertical component of the Earth's magnetic flux density is 5×10^{-5} T. What is the e.m.f. induced across the wing-tips of the plane?



We can treat the wingspan as a straight conductor, moving through the magnetic flux as shown in the diagram:

In each second the conductor moves 200 m and 'sweeps out' an area ΔA through the flux, where $\Delta A = 40\ \text{m} \times 200\ \text{m}$.

$$\textbf{e.m.f.,}\ \epsilon = N\frac{\Delta\Phi}{\Delta t} \qquad \text{where } N = 1 \text{ and } \Delta\Phi = B\,\Delta A$$

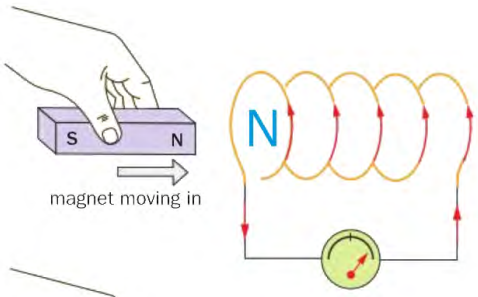
$$= \frac{B\Delta A}{\Delta t}$$

$$= \frac{5 \times 10^{-5}\ \text{T} \times (40\ \text{m} \times 200\ \text{m})}{1\ \text{s}} = \underline{0.4\ \text{V}} \quad (1\ \text{s.f.})$$

Lenz's Law

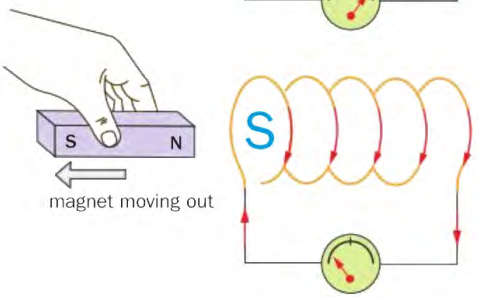
Faraday's Law tells us the size of the induced e.m.f., but we can find its direction using **Lenz's Law**:

The direction of the induced e.m.f. is such that it will try to oppose the change in flux that is producing it.



Lenz's Law is illustrated in the diagrams:

As you move the N-pole ***into*** the coil, an e.m.f is induced which drives a current round the circuit as shown.
Now use the right-hand grip rule (from page 259).
Can you see that the current produces a magnetic field with an N-pole at the end of the coil nearest to the magnet?
So the coil ***repels*** the incoming magnet, and in this way the induced current ***opposes*** the change in flux.



Why is the current reversed as you move the N-pole ***out***?
By Lenz's Law, the coil needs to attract the receding N-pole.

Lenz's Law is a result of the conservation of energy. If you move the magnet into the coil, you feel the repulsive force. You have to ***do work*** to move the magnet against this force. And so energy is transferred from you (or the system that is moving the magnet) to the electrical energy of the current.

You will often see Faraday's Law written in equation form like this: $\epsilon = -N\frac{\Delta\Phi}{\Delta t}$

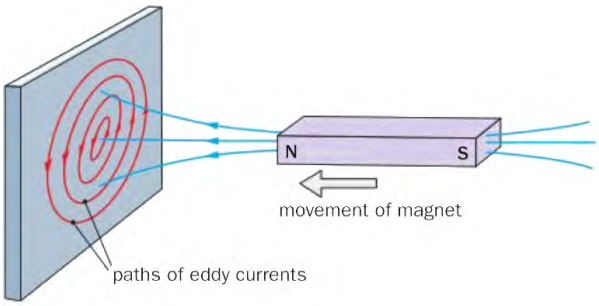
The minus sign reminds us that the e.m.f. is always induced in a direction so as to ***oppose*** the change in flux.



Phiz dreams of creating energy without work.

Eddy currents

What do you think happens in a piece of metal when it is moved in a magnetic field, or placed in a changing magnetic field?
Currents, called **eddy currents**, are induced in the metal. The diagram shows a simple example:

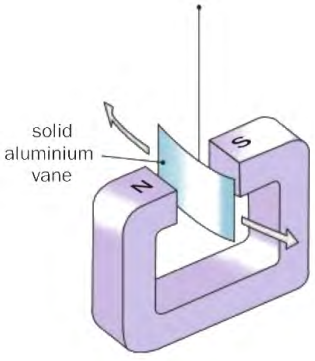


Eddy currents are always induced in a direction so as to ***oppose*** the motion or the change in flux producing them.

Eddy currents can be large, because the paths that they follow within the metal often have a very low resistance. These large currents can cause the metal to become very hot.

Magnetic braking

Why does this aluminium vane quickly stop oscillating when you swing it between the poles of the horseshoe magnet?



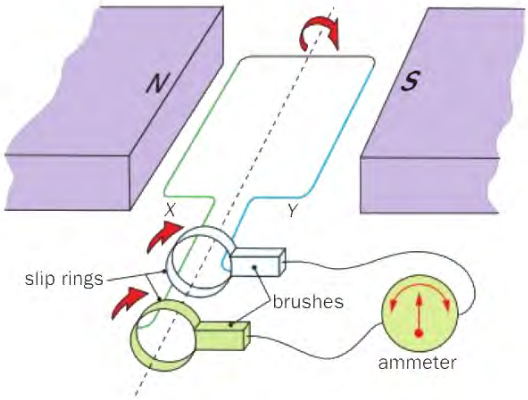
As it swings, the vane cuts through the lines of flux of the magnet, and so eddy currents are induced in the aluminium. The eddy currents then produce their own magnetic field. This field interacts with the field of the magnet, and produces a force to ***oppose*** the motion of the pendulum (Lenz's Law).

This effect is often called **magnetic braking** (see also page 278).

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The a.c. generator

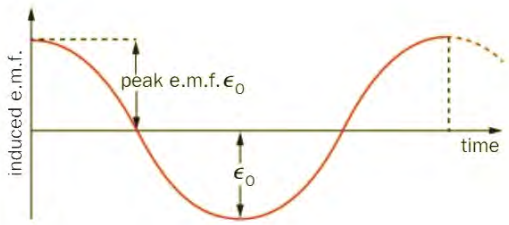
A generator converts ***kinetic energy into electrical energy***.
It consists of a rectangular coil that rotates in a magnetic field:



Each end of the coil is connected to a ***slip-ring***. The slip-rings rotate with the coil and press against stationary carbon brushes. Each side of the coil always makes contact with the same brush. This is shown by the blue and green shading on the diagram.

What happens if you rotate the coil at a steady rate?
As it turns the flux linkage of the coil constantly changes and the pointer on the ammeter swings from side to side, showing that alternating current (a.c.) is induced.

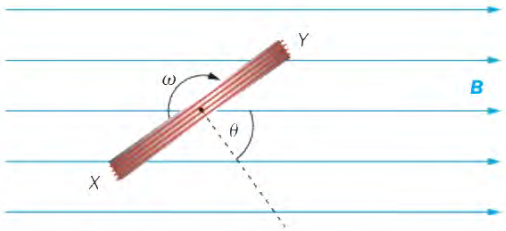
The graph shows how the size of the induced e.m.f. changes as the coil rotates once in the magnetic field:
One complete revolution of the coil gives one cycle of a.c.



How can we increase the size of the peak e.m.f. ϵ_0?
We must increase the rate of change of flux linkage of the coil as it spins in the magnetic field, by:

- using a coil with more turns N,
- using a coil with a larger cross-sectional area A,
- increasing the strength of the magnetic field B, or
- increasing the frequency of rotation of the coil, f.

Why is the graph of induced e.m.f. against time sinusoidal?
The diagram shows the coil end-on. It is rotating about its horizontal axis with angular speed ω (see page 79).
The flux linkage of the coil must be: $N\Phi = NBA\cos\theta$, where N is the number of turns and A is the area of the coil.



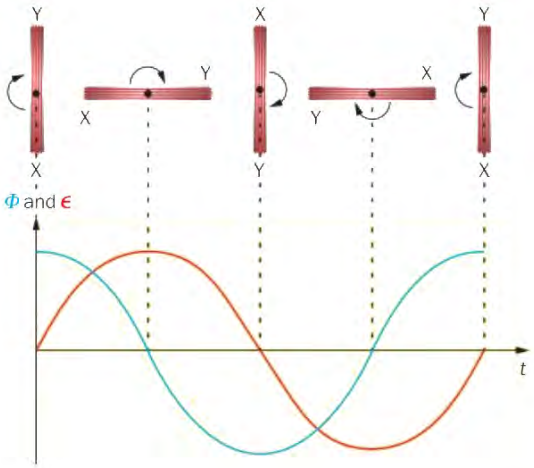
We start the time when $\theta = 0$; then after a time t the coil will have turned through an angle θ, where $\theta = \omega t$.
This means that the flux linkage of the coil is: $NBA\cos\omega t$.

Applying Faraday's Law (page 274) to the flux linkage equation, we get:

Induced e.m.f., $\epsilon = NBA\omega\sin\omega t$ or **$\epsilon = BAN\omega\sin\omega t$**

The graph shows how the flux Φ and the e.m.f. ϵ vary as the coil rotates. Notice that:

- the induced e.m.f. is zero when the flux linkage is a maximum,
- the e.m.f. is a maximum when the flux linkage is zero.



Can we explain these results?
Faraday's Law tells us that the magnitude of the induced e.m.f. is equal to the rate of change of flux linkage.
When Φ is a maximum, the gradient of the Φ curve is zero. There is no rate of change of flux linkage and no induced e.m.f.

When Φ is zero, the gradient of the Φ curve is a maximum. The rate of change of flux linkage is a maximum and so is the e.m.f.

What is the maximum value ϵ_0 of the induced e.m.f.?
The maximum value that the sine function can have is 1, and so:

$\epsilon_0 = BAN\omega$ or $\epsilon_0 = BAN \times 2\pi f$ (because $\omega = 2\pi f$)

Physics at work: Using electromagnetic induction

The metal detector

How does Phiz's metal detector work?
A.c. flows through the search coil in the search head, and this produces a constantly changing magnetic field around the coil.

What happens if there is a metal object in this changing field?
Eddy currents are induced in the metal object. This means that energy is transferred from the search coil to the metal object, and the current in the search coil changes. This is detected by the control box, which produces an audible signal.

How effective is the metal detector?
This depends on the size and the orientation of the metal object and how far below the ground it is buried.



Traffic lights are triggered in a similar way.
There is an induction loop in the road surface, and this creates an alternating magnetic field (like that of the search coil).
A car passing over the loop is detected, because it changes the current in the loop – just like a metal object in the metal detector.



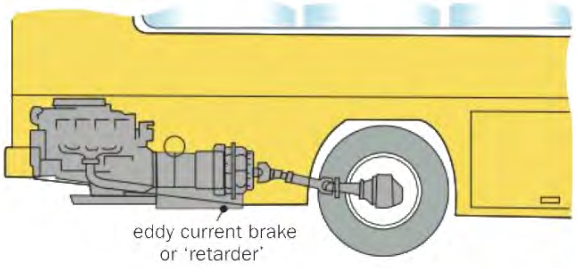
Induction heating

A ceramic hob is a type of electric cooker. It consists of three or four flat coils embedded in a ceramic surface.
Each coil is supplied with high-frequency a.c. – about 18 kHz.

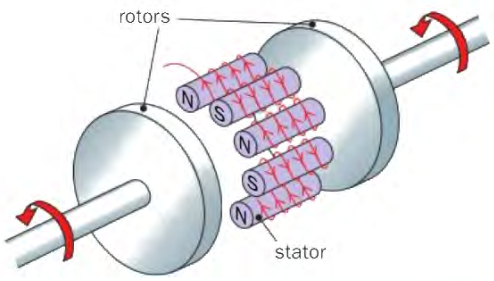
What happens when a metal pan is placed over one of the coils?
Eddy currents are induced in the base and the sides of the pan. As these currents flow through the metal pan, it becomes hot and so the food in the pan is cooked.
Can you explain why the ceramic surface stays relatively cool?

Electromagnetic brakes

Eddy-current brakes are often fitted to coaches and lorries.
Two discs (called rotors) are fixed to the wheel-shaft, so that they spin as the wheels turn.
Can you see the stator, fixed in place between the rotors?
The stator is an electromagnet made up of a number of coils.



What do you notice about the way the coils are wound?
When the driver activates the brake, a current is passed through the coils so that adjacent coils produce reversed magnetic fields.
The spinning rotors cut through the magnetic flux, and so eddy currents are induced in the discs.
The eddy currents flow so as to ***oppose*** the change producing them; hence the braking effect on the rotors and the wheels.



Why are the brakes not effective at slow speeds?
The discs spin slowly and the rate of flux cutting is reduced.
The smaller eddy currents produce a smaller braking effect.
Contact brakes (see page 30) are still needed for the final braking at slow speed.

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Summary

The magnetic flux Φ linking an area A when the perpendicular to the area makes an angle θ to a field of flux density B is: $\Phi = BA\cos\theta$

For a coil with N turns, total flux linkage $= NBA\cos\theta$

An e.m.f. is induced when the magnetic flux linkage changes, because either the flux density changes or the area linked by the flux changes.

Faraday's Law gives us the ***size*** of the induced e.m.f.
It states that the e.m.f. is equal to the ***rate*** of change of flux linkage:

$$\epsilon = N\frac{\Delta\Phi}{\Delta t}$$

Lenz's Law gives us the ***direction*** of the induced e.m.f. It states that the e.m.f. is induced so as to try to ***oppose*** the change that is causing it.

The e.m.f. in a coil rotating in a uniform field B is $\epsilon = BAN\omega\sin\omega t$

Questions

1. Can you complete these sentences?
a) When the magnetic ..... linking a coil changes, an e.m.f. is .....
b) Faraday's Law states that the ..... of the induced ..... is equal to the ..... of change of ..... linkage.
c) Lenz's Law states that the ..... of the induced ..... is always so as to ..... the change causing it.
d) When a magnet moves towards a coil, the size of the induced e.m.f depends on the ..... of the movement, the ..... of the magnet, and the cross-sectional ..... and number of ..... on the coil.

2. Explain the following:
a) No e.m.f. is induced when a straight wire is moved parallel to lines of magnetic flux.
b) A brass ring set to swing in a magnetic field quickly comes to rest; a plastic ring swings for much longer.
c) When a magnet is dropped through a plastic pipe it accelerates towards the ground; if dropped down an identical sized copper pipe it falls more slowly.

3. In the U.K. the vertical component of the Earth's magnetic field is 5.0×10^{-5} T. If the area of a village is 2.2×10^{6} m^2, how much flux passes through it?

4. A rectangular coil measures 5.0 cm by 8.0 cm. It is placed at 90° to a magnetic field of flux density 1.5 T. What is the flux linkage of the coil if it has 250 turns?

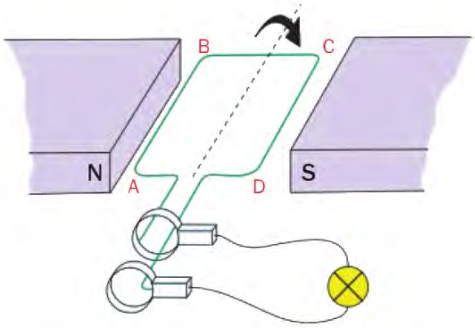
5. A search coil has 3400 turns and an area of 1.0 cm^2. It is placed between the poles of a magnet where the flux density is 0.40 T.
What is the average e.m.f. induced across the coil when it is snatched out of the field in 0.20 s?

6. A search coil with 2000 turns and an area of 1.5 cm^2 is used to measure the flux density between the poles of a horseshoe magnet. The coil is pulled out of the field in 0.30 s and the induced e.m.f. is 0.50 V.
What is the flux density of the magnet?

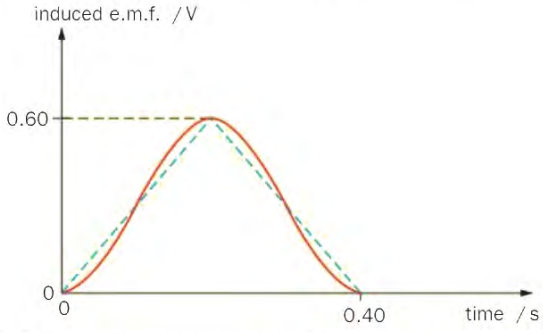
7. A wire of length 10.0 cm is moved downwards at a speed of 4.0 cm s^{-1} through a horizontal field of flux density 2.0 T. What is the e.m.f. induced in the wire?

8. An aircraft of wingspan 40 m flies horizontally at a speed of 250 m s^{-1}. An e.m.f. of 0.2 V is induced across the tips of the wings. What is the vertical component of the Earth's magnetic flux density?

9. The diagram shows a coil being turned in a horizontal magnetic field of flux density 50 mT.
a) In the position shown, does the induced current flow in the direction ABCD or DCBA?
b) The coil has 300 turns, area 15 cm^2 and rotates 10 times per second. What is the peak e.m.f.?



10. A search coil with 2500 turns and area 1.5 cm^2 is connected to a data-logger, placed between the poles of a magnet and quickly withdrawn.
a) Use the graph shown below to estimate the total change in flux linkage of the coil. (You can treat the graph as a triangle, as shown.)
b) Calculate the flux density of the magnet.



Further questions on page 316.